

Cross-supplier state-of-health forecasting for second-life LFP cells with a PyBaMM-trained encoder–decoder neural operator

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Second-life reuse of retired lithium iron phosphate (LFP) cells is limited by the time and cost of characterising each cell's future ageing. Conventional lifetime tests take months, and empirical models transfer poorly between suppliers with different chemistries. We present a physics-conditioned encoder–decoder neural operator that forecasts the full state-of-health (SoH) trajectory to end-of-second-life from a compact diagnostic snapshot plus ~ 50 cycles of field cycling data.

The workflow (Figure 1) has three stages. Phase 1 identifies beginning-of-life electrochemistry per anchor cell from OCV, HPPC, GITT and RPT. Phase 2 fits five degradation parameters — SEI growth rate, SEI molar volume, solvent diffusivity, lithium plating rate and negative-electrode loss of active material — by differential evolution against measured cycling. Phase 3 runs PyBaMM [1] with the SPM_e formulation and Prada2013 LFP chemistry [2] to synthesise 490 perturbed trajectories around seven anchor cells across three suppliers (CALB, EVE, REPT); a SoH-translation augmentation expands this into 14,096 (context, target) pairs covering the second-life SoH regime, teaching the operator to be translation-invariant on the absolute SoH axis. The operator is a θ -conditioned Neural ODE [3] whose decoder is integrated by `torchdiffeq` (dopri5) under a softplus-monotonic constraint, and whose context encoder ingests $K=50$ observed cycles of the target cell. Inference therefore consumes three inputs: the nearest-supplier θ vector as a physical prior, DCIR and initial-capacity features from a short characterisation, and the K observed cycles.

Held-out evaluation used one previously unseen cell per supplier (CALB_0029, EVE_0003, REPT_0031). Per-cell SoH root-mean-square errors were 0.95, 0.83 and 0.82 percentage points respectively, with mean 0.87 pp and cross-supplier standard deviation 0.06 pp; every held-out cell fell below one percentage point. Against the v6 single-shot baseline (mean 1.50 pp, std 0.32 pp) the encoder–decoder cuts mean error by 42% and cross-supplier spread by more than five-fold. Each forecast is a genuine forward projection: the decoder extends roughly 500 cycles beyond the observed window into deeper second-life SoH values. Limitations: the cross-supplier claim rests on three held-out cells; the forecast horizon is capped by the 2500-cycle PyBaMM corpus; and beyond-window projections remain experimentally unverified.

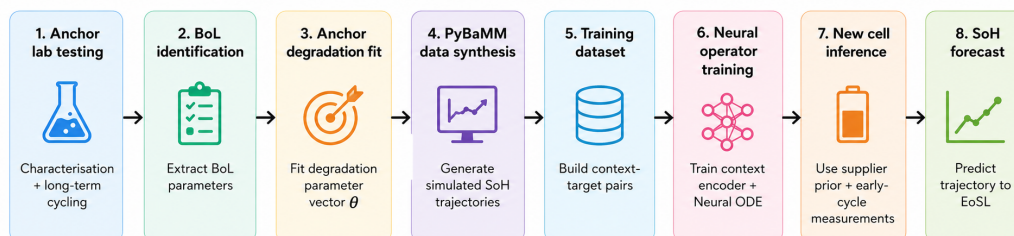


Figure 1: Pipeline: characterisation and long-term cycling feed Phase 1/2 identification; PyBaMM synthesises a perturbed training corpus; the SciML operator ingests θ , DCIR-derived features and $K=50$ observed cycles, and outputs the full SoH trajectory to end-of-second-life. Held-out cross-supplier RMSE: CALB_0029 0.95, EVE_0003 0.83, REPT_0031 0.82 percentage points (mean 0.87, std 0.06).

References:

1. V. Sulzer, S. G. Marquis, R. Timms, M. Robinson, S. J. Chapman, “Python Battery Mathematical Modelling (PyBaMM)”, *J. Open Res. Software* **9** (2021), 14.
2. M. Prada, D. Di Domenico, Y. Goff, J. Bernard, Y. Saurant-Mourat, F. Lhuet, “Simplified electrochemical and